

* understanding AI problem.

Ques: An AI problem is a task defined by an initial situation, a goal, and the actions move between them.

Types of AI problems

- 1) Ignorable.
- 2) Reversible
- 3) Irreversible
- 4) Toy vs Real.

Problem Formulation in AI.

- 1) Initial states
- 2) operators / Actions
- 3) Goal Test.
- 4) Path Cost.

Introduction to

Heuristic function

A heuristic - use evaluation.
apply generate - and - test, hill - climbing
and best - first Search.

Breadth - First Search (BFS)

~~Hill Climbing Search~~

→ A local Search that always moves to the ~~neighbour~~ with the best heuristic value - Climbing towards a peak.

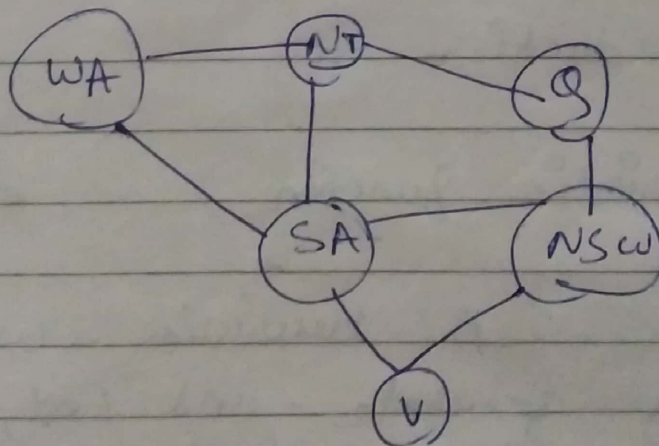
Problem Reduction and AND-OR graphs

Definition:-

Break a hard problem into simple Sub-problem Solved independently, then combine.

AND OR graphs.

Constraint Satisfaction Problems (CSP)



Means - Ends Analysis (MEA)

Definition.

A strategy that repeatedly reduces the difference between current state and the goal state.

Approaches to Knowledge Representation.

Syntax of Predicate Logic.

Constants Variables predicates.

Functions Connectives Quantifiers.

AI

Assignment - 2.

Translate into predicate logic

Let:

- $S(x)$ x is a student
- $P(x)$ x passed the exam.

Translate:

- a) All students passed the exam

Concept Learning:

Acquiring the definition of a general category from labelled examples of members (positive) and non-members (negative) of the category.

→ Target Concept $C: x \rightarrow (0, 1)$ - a boolean function over instances.

→ Learner sees training example $\{x, C(x)\}$.

→ Goal: hypothesis h such that

$$h(x) = C(x) \text{ for all } x.$$

→ It is a search for the "best" hypothesis fitting the data.

Running Example:-
Enjoy Sport.

Target: Enjoy Sport $\in \{Yes, No\}$.

Training examples for enjoy sport.

- Concept to be learned

- "Days on which Aldo enjoys the favorite water sport"

Attributes						$c(x)$
Sky	Temp	Humid	wind	water	Forecast	enjoy sport
Sunny	warm	Normal	strong	warm	Same	Yes
Sunny	warm	High	strong	warm	Same	Yes
Rainy	cold	High	strong	warm	change	No
Sunny	warm	High	strong	cool	change	Yes

1. Sky - Sunny/Rainy.

2. Temp - warm/cold

3. Humid - Normal/High.

4. wind - Strong

5. water: warm/cool.

6. Forecast: Same/change

$x_1 = (\text{Sunny, warm, Normal, strong, warm, Same})$

$x_2 = (\text{Sunny, warm, High, strong, warm, Same})$

$x_3 = (\text{Rainy, cold, High, strong, warm, change})$

$x_4 = (\text{Sunny, warm, High, strong, cool, change})$

$c(x_1) = \text{Yes}$

$c(x_2) = \text{Yes}$

$c(x_3) = \text{No}$

$c(x_4) = \text{Yes}$

$D = ((x_1, c(x_1)), (x_2, c(x_2)), (x_3, c(x_3)), (x_4, c(x_4)))$

Define $h(x)$ to find D_1 & D_2 .

$$D = (x_1, (x_1), (x_2, (x_2)), (x_3, (x_3)), (x_4, (x_4))).$$

$= [(\text{Sunny}, \text{warm}, \text{Normal}, \text{Strong}, \text{warm}, \text{same}), \text{Yes}], [(\text{Sunny}, \text{warm}, \text{High}, \text{Strong}, \text{warm}, \text{same}), \text{Yes}], [(\text{Rainy}, \text{Cold}, \text{High}, \text{Strong}, \text{warm}, \text{change}), \text{No}], [(\text{Sunny}, \text{warm}, \text{High}, \text{Strong}, \text{Cold}, \text{change}), \text{Yes}].$

To find hypothesis $h(x)$

D_1	D_2
Sunny, warm, Normal	Sunny, warm, High
Strong, warm, Same	Strong, warm, Same

$$h(x) = (\text{Sunny}, \text{warm}) = (x) = \text{Yes}$$

$$(x) = \text{No}$$

~~h~~
13/8

The Find-S Algorithm

Initialise h to the most specific hypothesis
 $h \leftarrow \{ \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \}$.

For each positive training instance x :

For each attribute constraint a in h :

If x satisfies $a \rightarrow$ do nothing; else
 replace a with the next more general
 constant value, or "?"

3. output hypothesis h .

① only positives

Negative examples never change h .

Buy Computer.

Examples.	Age	Income	Student	Credit Rating	Buy Computer
1	Youth	High	No	Fair	No
2	Middle Age	High	No	Fair	Yes
3	Senior	medium	No	Fair	Yes
4	Senior	low	yes	Fair	Yes
5.	Senior.	low	yes	Excellent	No.

Attributes.

1. Age - Youth / Middle Age / Senior.

2. Income - High / medium / low.

3. Student - Yes / No.

4. Credit Rating - Fair / Excellent.

$x_1 = (\text{Youth, High, No, fair}) \in (x_1) - \text{No}$

$x_2 = (\text{Middle Age, High, No, fair}) \in (x_2) - \text{Yes}$

$x_3 = (\text{Senior, middle, No, fair}) \in (x_3) - \text{Yes}$

$x_4 = (\text{Senior, low, Yes, Fair}) \in (x_4) - \text{Yes}$

$x_5 = (\text{Senior, low, Yes, Excellent}) \in (x_5) - \text{No}$

$h(x) = (\emptyset, \emptyset, \emptyset, \emptyset)$

Positive Examples $\rightarrow x_2, x_3, x_4$

$x_2, x_3 \rightarrow h(x) \Rightarrow (?, ?, \text{No, fair})$

$x_2, x_3 \text{ and } x_4 \Rightarrow h(x) \Rightarrow (?, ?, ?, \text{fair})$

-ve Example $\Rightarrow x_1, x_5$

$h(x) \Rightarrow (?, ?, ?, ?)$

→ supervised → Label data

unsupervised → unlabel data

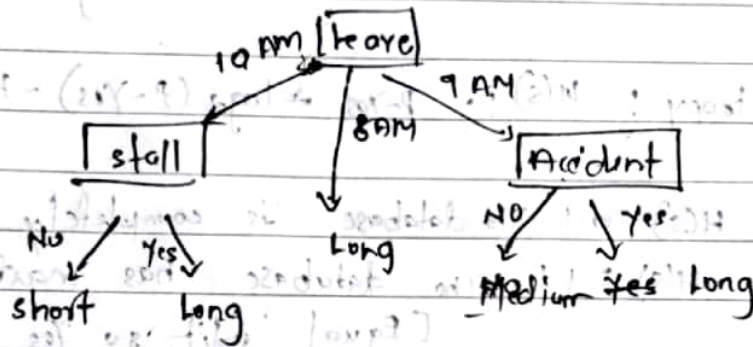
What is a Decision Tree?

- An inductive learning task

- A predictive model based on a branching series of

These smaller Boolean tests over Boolean tests less complex than a one-stage classifier.

Example :-



2. How to create a Decision Tree

→ first make a list of attributes that can measure
... must be discrete.

→ Then choose a target attribute that want to Predict

→

Algorithm (ID3)

→ choose the best attribute

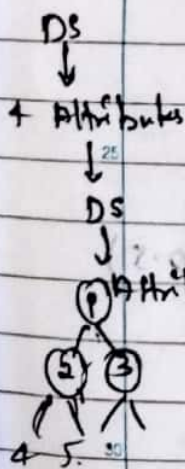
→ Repeat this process for recursively for each child

→ stop when

→ All the instances have the same target value

→ There is no more attributes

→ There are no more instances.



Day	outlook	Humidity	wind	Play & Tennis (or)
D1	sunny	High	Weak	No
D2	sunny	High	strong	No
D3	overcast	High	Weak	Yes
D4	Rainy	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	strong	No

Entropy: $H(S) = -P_{-yes} \times \log_2(P_{-yes}) - P_{-no} \times \log_2(P_{-no})$

$H(S) = 0$: The database is completely pure

$H(S) = 1$: The database has maximum impurity
[Equal split so Yes & No]

Information Gain $IG(S, A)$

Details Steps

Step 1: calculate Total Entropy of Dataset $H(S)$

Total NO: of dataset $N=6$

Target = Yes: 3 (D3, D4, D5)

Target = NO: 3 (D1, D2, D6)

Probability

$$P_{-yes} = P_{-NO} = 3/6 = 0.5$$

$$H(S) = 0.5 \times \log_2(0.5) + 0.5 \times \log_2(0.5)$$

$$= 0.5 \times (-1) + 0.5 \times (-1)$$

$$= 0.5 + 0.5$$

$$= 1.0$$

Step 2: calculate information Gain by for Root section

Attribute A: outlook

outlook takes three distinct values: sunny, overcast, and rain.

→ Sunny (2 instance: D_1, D_2): Target = [No, No] [0 Yes, 2 No]
 $H(\text{sunny}) = 0$ (Pure Node)

→ Overcast (1 instance: D_3): Target = [Yes] (Yes, 0 No)

→ Rain (3 instance: D_4, D_5, D_6): Target [Yes, Yes, No]
 [2 Yes, 1 No]

$$P(\text{Yes}) = \frac{0+0+2}{6} = \frac{2}{6} = 0.333 \quad P(\text{No}) = \frac{2}{6} = 0.333$$

$$H(\text{Rain}) = -(0.667) \times \log_2(0.667) - 0.333 \times \log_2(0.333)$$

$$= -(0.667) \times (-0.585) - 0.333 \times (-1.585)$$

$$= 0.1390 + 0.528 = 0.667$$

Weighted Entropy for outlook

$$E(\text{outlook}) = \left(\frac{2}{6}\right) \times H(\text{sunny}) + \left(\frac{1}{6}\right) \times H(\text{overcast}) + \left(\frac{3}{6}\right) \times H(\text{rain})$$

$$E(\text{outlook}) = \left(\frac{2}{6}\right) \times (0) + \left(\frac{1}{6}\right) \times (0) + \left(\frac{3}{6}\right) \times (0.667)$$

$$= 0 + 0 + 0.333$$

$$\text{Weighted Entropy} = 0.333$$

$$IG(S, A) = IG(S, \text{outlook})$$

$$IG(S, \text{outlook}) = H(S) - E(\text{outlook})$$

$$H(S) = 0.918$$

$$IG(S, \text{outlook}) = 0.918 - 0.333 = 0.585$$

$$IG(S, \text{outlook}) = 0.585$$

Attribute B: Humidity

Humidity takes two distinct values: High and Normal

→ High (4 instances: D_1, D_2, D_3, D_4) : Target = [No, Yes, Yes, No]
(2 Yes, 2 No)

$$P_{\text{Yes}} = \frac{2}{4} = 0.5, \quad P_{\text{No}} = \frac{2}{4} = 0.5 \Rightarrow H(\text{High}) = 1$$

→ Normal (2 instances: D_5, D_6) : Target = [Yes, No] (1 Yes, 1 No)
 $P_{\text{Yes}} = \frac{1}{2} = 0.5, \quad P_{\text{No}} = \frac{1}{2} = 0.5 \Rightarrow H(\text{Normal}) = 1$

→ Weighted Entropy for humidity

$$E_{\text{weighted}}(\text{Humidity}) = \left(\frac{6}{10}\right) * (1.00) + \left(\frac{2}{10}\right) * (1.00) = 1.00$$

$$IG(S, \text{Humidity}) = H(S) - E_{\text{weighted}}(\text{Humidity}) = 1.0 - 1.0 = 0.00$$

Attribute C: wind

wind takes two distinct values: weak and strong

→ weak (4 instances: D_1, D_3, D_4, D_5) : Target = [No, Yes, Yes, No]

$$P_{\text{Yes}} = \frac{3}{4} = 0.75; \quad D_2, D_6 : \text{Target} = [\text{No}, \text{No}]$$

(0 Yes, 2 No)

$$H(\text{strong}) = 0 \quad \text{Pure Node}$$

Weighted Entropy for wind:

$$E_{\text{weighted}}(\text{wind}) = \left(\frac{6}{10}\right) * (0.811) + \left(\frac{2}{10}\right) * (0) = 0.541$$

$$IG(S, \text{wind}) = H(S) - E_{\text{weighted}}(\text{wind}) = 1.0 - 0.541 = 0.459$$

step 3 : Root Node selection

Attribute	Weighted Entropy	Weighted	Information Gain
outlook	0.459		0.541 (max)
humidity	1.000		0.00
wind	0.541		0.459

Decision : Since outlook has the high Information Gain.

step 4 : splitting the Rain subset (Detailed math)

subset S-Rain contains D_4, D_5 and D_6

Day,	Humidity	wind	play Tennis?
D_4	High	Weak	Yes
D_5	Normal	Weak	Yes
D_6	Normal	Strong	No

Current Entropy of Rain subset : $H(S-Rain) = 0.918$
 Remaining Candidate Attribute : Humidity, wind.

A. Candidate Attribute : Humidity (in Rain subset)
 High (1 instance : D_4) : Target = [Yes] $\Rightarrow H(High) = 0$

$$P_{\text{yes}} = \frac{1}{2} = 0.5, \quad P_{\text{no}} = \frac{1}{2} = 0.5 \Rightarrow H(Normal) = 1.0$$

$$IG(S-Rain - humidity) = 0.918 - 0.667 = 0.251$$